

SOURAV KUMAR

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PROFESSIONAL SUMMARY

AI-focused software engineer and Computer Science undergraduate who designs and ships production LLM systems end to end, spanning retrieval-augmented generation and agentic tool-calling architectures. Experienced with OpenAI GPT-4o function calling, Model Context Protocol (MCP), prompt engineering and multi-tenant AI application design, on a strong TypeScript, Node.js and PostgreSQL foundation. Seeking an AI Engineer role focused on Generative AI.

TECHNICAL SKILLS

Generative AI & LLM: OpenAI API (GPT-4o, text-embedding-3), Retrieval-Augmented Generation (RAG), Vector Embeddings & Semantic Search, Function & Tool Calling, Model Context Protocol (MCP), Agentic Orchestration, Prompt Engineering, Structured Output (JSON Schema)

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Backend & APIs: Node.js, Express.js, tRPC, REST & OpenAPI, OAuth 2.0, JWT, Server-Sent Events, Webhooks, Multi-Tenant Architecture

Frontend: React.js, Next.js, Tailwind CSS, shadcn/ui

Databases: PostgreSQL, pgvector, Drizzle ORM, MongoDB, Mongoose ODM, SQL

Cloud & Tools: Docker, Turborepo, pnpm, Git, GitHub, Linux

Foundations: Data Structures & Algorithms, System Design, Agile

PROJECTS

StudyBookLM — *NotebookLM-Style RAG Document Intelligence Platform* 2026

- Built a retrieval-augmented generation platform that turns uploaded study material into a conversational knowledge base, answering questions with OpenAI GPT-4o grounded in the user's own documents.
- Implemented the full ingestion and retrieval pipeline in TypeScript: document chunking, embedding generation with OpenAI text-embedding-3, and vector similarity search over pgvector in PostgreSQL.
- Engineered multi-document retrieval so a single query gathers and synthesises context across an entire uploaded corpus rather than one file at a time.
- Extended the pipeline beyond Q&A to generate derived study artefacts — summaries, study guides and mind maps — from retrieved source passages.

Tech: TypeScript, Node.js, OpenAI GPT-4o, text-embedding-3, PostgreSQL, pgvector, RAG

Ultrahuman — *Agentic AI Assistant for Gmail & Google Calendar* 2026

- Built a multi-tenant agentic AI assistant that executes real Gmail and Google Calendar actions from natural language, driving an OpenAI GPT-4o tool-calling loop over Model Context Protocol (MCP) tools.
- Implemented the agent orchestration loop with bounded multi-round tool execution, feeding tool results and captured errors back into the model so it recovers from failed calls mid-task instead of aborting.
- Wrote a Zod-to-JSON-Schema compiler that converts MCP tool definitions into OpenAI function schemas, keeping tool contracts type-safe and single-sourced across the monorepo.
- Enforced per-user tool isolation using tenant-scoped clients and envelope-encrypted OAuth credentials, so the agent can only ever act on the signed-in user's own mailbox and calendar.
- Shipped realtime inbox and calendar updates over a webhook → SSE → query-invalidation pipeline, with typed tRPC routes auto-published as OpenAPI docs.

Tech: TypeScript, OpenAI GPT-4o, MCP, tRPC v11, Next.js, Express, PostgreSQL, Drizzle ORM, Turborepo, Google OAuth 2.0, Docker

EDUCATION

Raj Kumar Goel Institute of Technology — B.Tech, Computer Science & Engineering Sep 2024 – Aug 2028

Ghaziabad, India | Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Chaicode Academy — Full-Stack JavaScript (MERN) Bootcamp, Online Jan 2026 – Present

ACHIEVEMENTS

- Third Prize, Changeathon — National Social Summit 2026, IIT Roorkee (national-level tech-for-social-impact hackathon): built an AI-powered surveillance system with real-time threat detection and intelligent monitoring, proposed to the Government of India.
- Actively solving Data Structures & Algorithms problems on LeetCode in preparation for technical interviews.